Velocity and position graphs

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a position-time graph to velocity using slope.

SUCCESS CRITERIA - I can...

1. Read positions and elapsed times from a graph.
2. Find velocity from position change divided by time.
3. Explain the difference between speed and velocity.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Start at 3 m, move to 10 m, then finish at 5 m. Draw the route. Find distance, displacement with direction and final position.

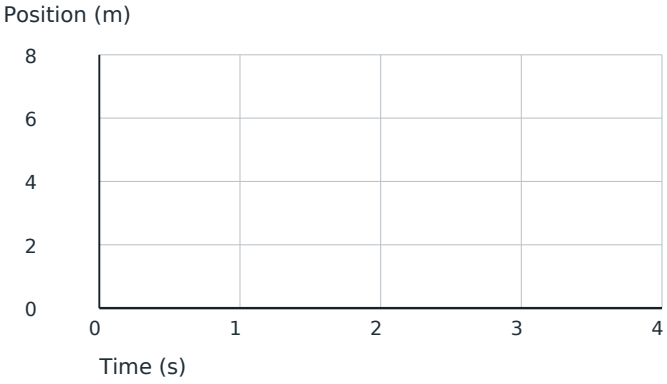
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_{avg} = \Delta x/\Delta t$; average speed = distance/Δt. Δ means change; Δt is elapsed time. Position-time slope gives velocity.

Worked reasoning: A cart moves from 1 m to 7 m in 3 s at steady velocity. The position-time slope is (7-1)/3=2 m/s.

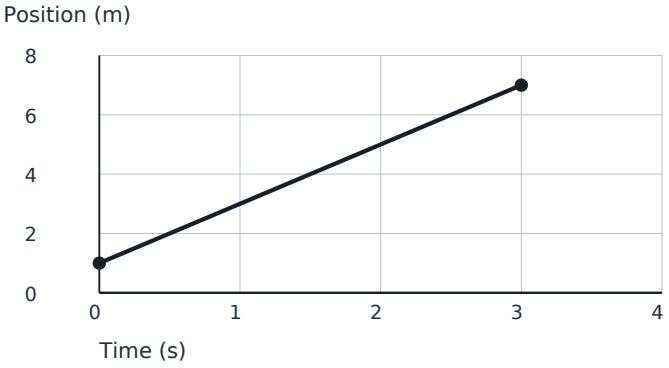
Your turn: Use the blank axes to plot the cart's two measured positions and connect them. Draw a slope triangle. Label its rise in meters and its run in seconds.



Marzano check - circle: ① ② ③

03 Read the motion picture

A motion sensor tracks a cart on a straight test track. Read its position at 0 s and 3 s. Which direction is it moving? Use the changing position to explain your answer.



Marzano check - circle: ① ② ③

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Continue down each column.

04 Find the rate

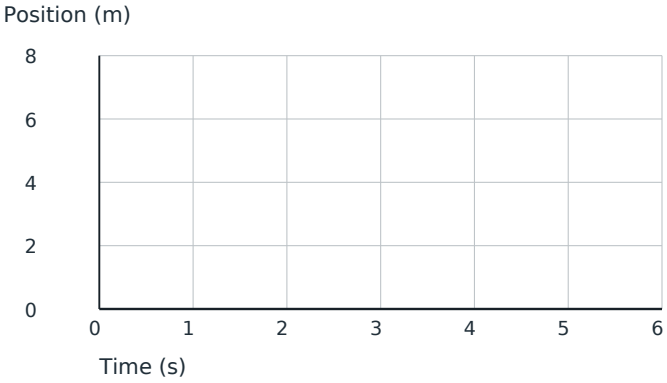
Use the graph in 3. Position change = ____ - ____ = ____ m. Time = ____ s. Velocity = ____ m/s to the ____.

Marzano check - circle: 1 2 3

05 A pause

During another trial, the cart's position stays at 5 m from 4 s to 6 s. Sketch that interval on the axes. Is the cart moving? Explain what stays unchanged.

Support: The cart is ____ because its position ____.

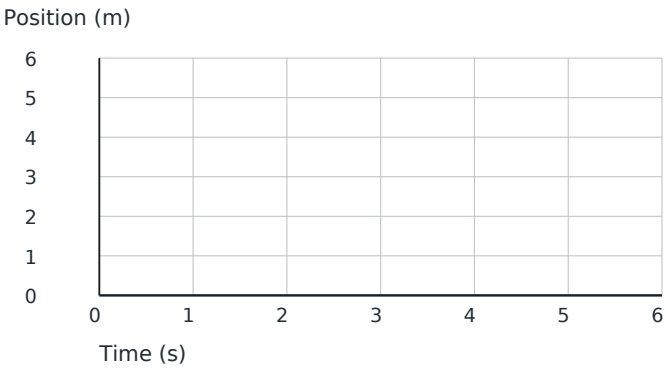


Marzano check - circle: 1 2 3

06 Draw a walk with a pause

A student walks along a marked hallway. Plot (time, position): (0 s, 1 m), (2 s, 5 m), (4 s, 5 m), (6 s, 1 m). Connect neighboring points with straight lines. Label each interval: moving right, stopped, or moving left.

Support: Data: t = 0, 2, 4, 6 s; x = 1, 5, 5, 1 m.



Marzano check - circle: 1 2 3

07 Distance or displacement?

For the walk in 6, find distance traveled and displacement. Why is average velocity zero even though the person walked?

Marzano check - circle: 1 2 3

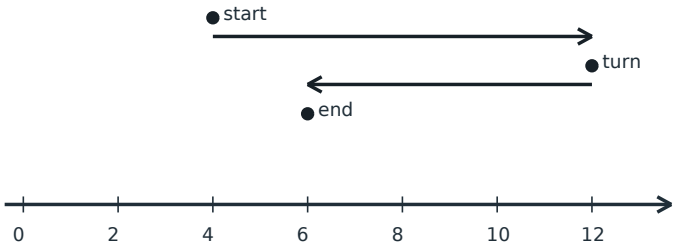
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Continue down each column.

08 Review: connect an earlier idea

Start at 4 m, move to 12 m, then finish at 6 m. Draw the route. Find distance, displacement with direction and final position.

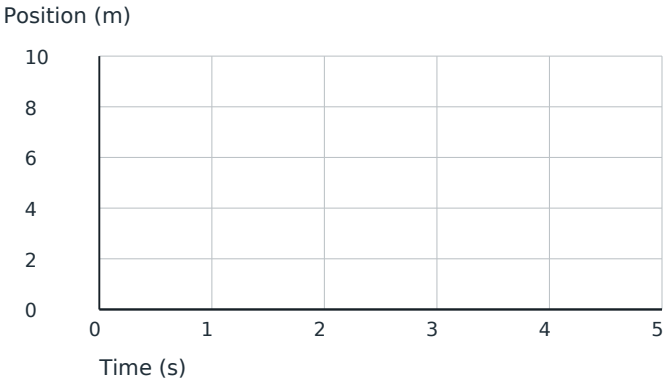


Position (m); right is positive

Marzano check - circle: 1 2 3

09 Exit check

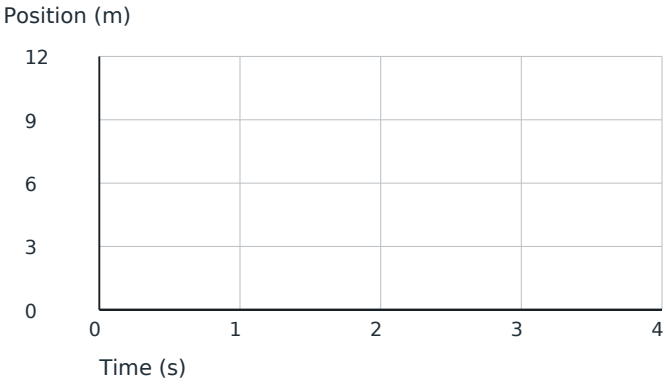
A remote-control car moves steadily from the 9 m mark to the 3 m mark in 5 s. Find its velocity, including direction, and draw its position-time graph.



Marzano check - circle: 1 2 3

10 STRETCH: test the model

Draw two straight position-time graphs with different starting positions but the same velocity of 2 m/s. Explain the feature they share.



Marzano check - circle: 1 2 3